

**AMENDMENT NO. 2 JULY 2008
TO
IS 5301 : 1987 SPECIFICATION FOR
SODIUM CHLORATE**

(First Revision)

(Page 6, Appendix A) — Substitute the following for the existing

A-2 SODIUM CHLORATE

A-2.0 Outline of the Method — Sodium chlorate is determined by estimating the amount of standardized ferrous ammonium sulphate consumed by the material after reduction

A-2.1 Reagents

A-2.1.1 0.1 N Potassium Dichromate Solution — Dry Potassium Dichromate (AR grade) at 140°-150°C for 30-60 min and cool in a closed vessel in a desiccator. Dissolve 4.904 g in water and dilute in a volumetric flask to 1 000 ml.

A-2.1.2 0.2 N Ferrous Ammonium Sulphate Solution in 4 N Sulphuric Acid — Dissolve 78.4 g of ferrous ammonium sulphate (AR grade) ($\text{FeSO}_4(\text{NH}_4)_2 \cdot \text{SO}_4 \cdot 6\text{H}_2\text{O}$, M wt-392) in 1 000 ml of 4 N sulphuric acid.

A-2.1.3 1:1 Ortho Phosphoric Acid Solution — Take 100 ml of ortho phosphoric acid and dilute with 100 ml of water.

A-2.1.4 Sodium Diphenyl Amine Sulphonate Indicator (0.2%) — Dissolve 0.20 g of sodium diphenyl amine sulphonate in 100 ml of distilled water.

A-2.2 Procedure — Dry the material over sulphuric acid for 24 h. Weigh accurately about 2.13 g (M) of sodium chlorate. Dissolve it in approximately 100 ml of water in a volumetric flask and make up the volume to one litre (0.02M). In a 250-ml conical flask, place 25 ml of above sodium chlorate solution, 25 ml of 0.2 N ferrous ammonium sulphate solution in 4 N sulphuric acid and add cautiously 12 ml of concentrated sulphuric acid. Heat the mixture to boiling (in order to ensure completion of reduction) and cool to room temperature by placing the flask under running tap water. Add 20 ml of 1:1 ortho phosphoric acid followed by 0.5 ml of sodium diphenyl amine sulphonate

Amend No. 2 to IS 5301 : 1987

indicator Titrate the excess of ferrous ion with standard 0.1 N potassium dichromate solution to a first tinge of purple coloration, which remain on stirring. Let the reading be V_1

Perform blank titration with the ferrous ammonium sulphate solution by repeating the procedure but using 25 ml of distilled water in place of sodium chlorate solution. Let the reading be V_2 (Blank). The difference in reading is equivalent to the amount of sodium chlorate added.

A-2.3 Calculation

$$\text{Sodium chlorate, percent by mass} = \frac{(V_2 - V_1) \times N \times 70.96}{M}$$

where

N = normality of potassium dichromate solution used for titration,

M = mass of sodium chlorate sample in g dissolved per litre of solution,

V_2 = volume in ml of potassium dichromate solution consumed in blank experiment, and

V_1 = volume in ml of potassium dichromate solution consumed by excess of reducing agent

(CHD 01)

Reprography Unit, BIS, New Delhi, India